

COM S 2300 HW2

1 a) This statement basically means that all faculty members, say x, have asked questions to at least 2 different students, say y and z. So from this statement we can say that:

$$\forall x(F(x) \rightarrow \exists y \exists z(S(y) \wedge S(z) \wedge y \neq z \wedge A(x,y) \wedge A(x,z)))$$

b) This statement basically means that there are 2 faculty members, say y and z, who haven't asked any questions to any student, say x. So from this statement we can say that:

$$\exists y \exists z(F(y) \wedge F(z) \wedge y \neq z \wedge \forall x(S(x) \rightarrow \neg A(y,x)) \wedge \forall x(S(x) \rightarrow \neg A(z,x)))$$

2 a) S(x): x is a student in class.

P(x): x has visited

A(x): x has visited Arc de Triomphe

Goal: $\exists x(S(x) \wedge A(x))$

Given Statements:

- 1) $S(M) \wedge P(M)$ (Mary is student in the class and she has visited Paris)
- 2) $\forall x[P(x) \rightarrow A(x)]$ (Everyone who visits Paris also go to Arc De Triomphe)

Steps	Reason
$S(M) \wedge P(M)$	Premise 1
$P(M)$	Simplification
$P(M) \rightarrow A(M)$	Premise 2
$P(M) \text{ and } [P(M) \rightarrow A(M)] = A(M)$	Modus Ponens

Now, since Marry is a student in the class (given in the question) and she has visited Arc De Triomphe (proven from the table), we can say that someone in the class has visited Arc De Triomphe $[\exists x(S(x) \wedge A(x))]$, hence proved.

b) C(x): x is a college town

M(x): x is in the midwest.

F(x): x is fun

Goal: $\exists x(M(x) \wedge F(x))$

Given Statements:

- 1) $\exists x(C(x) \wedge M(x))$ (There are some colleges in the midwest)
- 2) $\forall x(C(x) \rightarrow F(x))$ (All college towns are fun)

Steps	Reason
$C(x) \wedge M(x)$	Premise 1
$C(x) \rightarrow F(x)$	Premise 2
$C(x)$	Simplification of premise 1
$C(x) \text{ and } (C(x) \rightarrow F(x)) = F(x)$	Modus ponens
$M(x)$	We know that $M(x)$ is true from statement 1 which says that there are some college towns in the Midwest.
$F(x) \text{ and } M(x) = (F(x) \wedge M(x))$	Conjunction

Therefore, we can say that there is a college town in the midwest that is fun to live in.

Hence proved.

- c) $H(x)$: x lives in Hawaii
 $O(x)$: x lives close to the ocean
 $S(x)$: x knows how to surf
 $B(x)$: x owns a surfboard

Goal: $\forall x(H(x) \rightarrow B(x))$

Given Statements:

- 1) $\forall x(H(x) \rightarrow O(x))$ (All people who live in Hawaii, live close to the ocean)
- 2) $\exists x(\neg S(x) \rightarrow \neg O(x))$ (A person who doesn't know how to surf, don't live close to the ocean)
- 3) $\forall x(S(x) \rightarrow B(x))$ (All people who know how to surf own a surfboard)

Steps	Reason
$H(x) \rightarrow O(x)$	Premise 1
$\neg S(x) \rightarrow \neg O(x)$	Premise 2
$S(x) \rightarrow B(x)$	Premise 3
$\neg S(x) \rightarrow \neg O(x) = O(x) \rightarrow S(x)$	Contrapositive of premise 2
$(H(x) \rightarrow O(x)) \text{ and } (O(x) \rightarrow S(x)) = (H(x) \rightarrow S(x))$	Hypothetical syllogism of premise 1 and contrapositive of premise 2
$(H(x) \rightarrow S(x)) \text{ and } (S(x) \rightarrow B(x)) = (H(x) \rightarrow B(x))$	Hypothetical syllogism of the argument above and premise 3.

Therefore, from the proof we can say that all people who live in Hawaii own a surfboard.

Hence proved.

3 a) $\text{right}(x,y) \equiv \text{left}(y,x)$

Reason: Since x is on the right side of y, so y will be on the left side of x.

b) $\text{onstage}(x) \equiv \exists y[\text{left}(x,y) \vee \text{left}(y,x)]$

Reason: Since x is on stage and there are at least 2 people, so it could either be on the right side of y or on its left side.

c) $\text{leftmost}(x) \equiv (\text{onstage}(x) \wedge \forall y(\neg \text{left}(y,x)))$

Reason: Since x is in line that means that it is on the stage and since it's the leftmost in the line so that means no one is on its left.

d) $z = \text{all people standing in the line}$

$\text{second}(x) \equiv \exists y \forall z (\text{leftmost}(y) \wedge \text{right}(z,x) \wedge z \neq y \wedge z \neq x)$

Reason: Since x is the second from the left in line, this means that there is a person towards its left, and all other people are standing towards its right in the line.

e) $z = \text{all people standing in the line}$

$\text{immediate-left}(x,y) \equiv \exists y \forall z [\text{left}(x,y) \wedge (\text{right}(z,y) \vee \text{left}(z,x)) \wedge z \neq y \wedge z \neq x]$

Reason: Since x is immediately to the left of y, that means that all other people are either on the left of x or at the right of y.

4) **Domain:** A set of all integers

Definitions:

$P(x) : x \text{ is odd}$

$Q(x)$: x is even

$\forall x(P(x) \rightarrow Q(x))$: This statement means that every integer x must be both odd and even at the same time or it should be neither one of them. This statement is false since x can only be even or odd at the same time.

$\forall xP(x) \longleftrightarrow \forall xQ(x)$:

- 1) $\forall xP(x)$: This statement means that all integers are odd. It is a false statement since we have even integers as well.
- 2) $\forall xQ(x)$: This statement means that all integers are even. It is a false statement since we have odd integers as well.

Since both $\forall xP(x)$ and $\forall xQ(x)$ are false, we can say that $\forall xP(x) \longleftrightarrow \forall xQ(x)$ returns true.

Also, since $\forall x(P(x) \rightarrow Q(x))$ returns false and $\forall xP(x) \longleftrightarrow \forall xQ(x)$ returns true, we can say that they are not equivalent logically.

We can take an example where $x = 2$ to provide a counterexample.

$\forall x(P(2) \rightarrow Q(2))$: This statement is false since 2 is just an even integer.

$\forall xP(2) \longleftrightarrow \forall xQ(2)$:

- 1) $P(2)$: This statement is false since we know that 2 is an even integer, so it defies the statement that all integers are odd.
- 2) $Q(2)$: This statement means that 2 is even. It is a true statement, however if we take $Q(3)$, this statement will become false since it is an odd integer.

Hence, by providing this counterexample we have proved that $\forall x(P(x) \rightarrow Q(x)) \neq \forall xP(x) \longleftrightarrow \forall xQ(x)$, since the former results in false and the latter returns a true value.